

Chemistry Source Material

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How does the pH scale work?

The pH scale measures how acidic or basic (alkaline) an aqueous solution is. It is defined as the negative base-10 logarithm of the hydrogen ion concentration: $\text{pH} = -\log_{10}[\text{H}^+]$. The scale conventionally runs from 0 to 14, although values outside that range are possible for very strong acids or bases.

A solution with a pH below 7 is acidic, meaning it has a higher concentration of free hydrogen ions (H^+) than pure water. A solution with a pH of exactly 7, such as pure water at 25 degrees Celsius, is neutral. A solution with a pH above 7 is basic, or alkaline, meaning it has a lower concentration of free hydrogen ions and a correspondingly higher concentration of hydroxide ions (OH^-).

Because the scale is logarithmic rather than linear, each single whole-number step represents a tenfold change in hydrogen ion concentration. A solution at pH 4 is ten times more acidic than a solution at pH 5, and one hundred times more acidic than a solution at pH 6. This is why small differences in a measured pH value can correspond to large differences in actual acidity.

Common examples help calibrate the scale: lemon juice has a pH around 2 (strongly acidic), black coffee is around 5 (mildly acidic), pure water is 7 (neutral), baking soda solution is around 9 (mildly basic), and household bleach or soap solutions can reach pH 12-13 (strongly basic).

pH is typically measured using a calibrated electronic pH meter (a glass electrode sensitive to hydrogen ion activity) or, less precisely, with colorimetric indicators such as litmus paper or universal indicator solution, which change color across the pH range.

Why do atoms form covalent bonds?

Atoms form covalent bonds because doing so lowers the total potential energy of the system, moving each atom's outer (valence) electron shell toward a more stable, lower-energy configuration -- for most main-group elements, a full outer shell of eight electrons (the octet rule), or two for hydrogen and helium.

A covalent bond forms when two atoms -- typically two nonmetals with similar electronegativity -- each contribute one electron to a shared pair. That shared pair of electrons occupies the region between the two

nuclei and is simultaneously attracted to both positively charged nuclei, which is the electrostatic force that holds the atoms together.

This is different from an atom simply losing or gaining an electron outright: in covalent bonding neither atom fully gives up its electron. Instead, the electrons are shared, which lets both atoms count the shared pair toward their own outer shell simultaneously, satisfying both atoms' drive toward a full shell without requiring a net charge transfer.

The number of covalent bonds an atom tends to form is governed by how many additional electrons it needs to complete its outer shell. For example, a hydrogen atom needs one more electron and typically forms one bond; an oxygen atom needs two more electrons and typically forms two bonds, as in water (H₂O).

The result of covalent bonding is a molecule: a discrete, electrically neutral unit in which specific atoms are held together by shared electron pairs, with a fixed, well-defined composition (for example, exactly two hydrogen atoms and one oxygen atom in every water molecule).

What is the difference between ionic and covalent bonding?

Ionic and covalent bonds are the two principal ways atoms combine to form compounds, and they differ in what happens to the electrons involved. In ionic bonding, one or more electrons are transferred completely from one atom to another. In covalent bonding, electrons are shared between atoms rather than transferred.

The deciding factor is usually the electronegativity difference between the two atoms -- how strongly each atom's nucleus attracts shared electrons. A large electronegativity difference, typically between a metal (which readily gives up electrons) and a nonmetal (which readily accepts them), favors ionic bonding. A small electronegativity difference, typically between two nonmetals, favors covalent bonding.

In ionic bonding, the atom that loses an electron becomes a positively charged ion (a cation), and the atom that gains it becomes a negatively charged ion (an anion). These oppositely charged ions attract each other electrostatically. Sodium chloride (table salt, NaCl) is a classic example: sodium transfers an electron to chlorine, forming Na⁺ and Cl⁻ ions.

The resulting structures also differ. Ionic compounds do not form individual molecules; instead, the oppositely charged ions arrange themselves into a repeating, three-dimensional crystal lattice held together by electrostatic attraction throughout the whole structure. Covalent compounds form discrete molecules, each with a fixed, specific number of atoms bonded by shared electron pairs.

These structural differences produce different bulk properties. Ionic compounds tend to have high melting points and conduct electricity when dissolved in water or melted, because the charged ions become free to move. Covalent compounds tend to have lower melting points and generally do not conduct electricity, because their molecules are electrically neutral overall.